JYOTHIR RAGHAVALU BHOGI

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Jyothir-dev | LinkedIn
Jyothir-dev (Jyothir) (github.com)

EDUCATION

Lovely Professional University — Jalandhar, India
Bachelor of Technology (Computer Science and Engineering) — **July 2022 - May 2026**

SKILLS SUMMARY

- **Programming**: Python, SQL
- **Machine Learning**: Machine Learning, Deep Learning, Model Evaluation, Performance Optimization
- **AI / NLP**: Natural Language Processing (NLP), Transformer Models (BERT, RoBERTa), Large Language Models (LLMs), Embeddings
- **AI Systems**: Semantic Search, Similarity Search, Retrieval Augmented Generation (RAG), Vector Databases, Vector Retrieval Systems
- **Frameworks**: PyTorch, TensorFlow
- **Data**: Data Processing, Feature Engineering, Dataset Preparation, Structured and Unstructured Data Handling
- **Deployment**: Docker, CI/CD
- **Version Control**: Git
- **Collaboration**: Agile, SDLC, Cross-Functional Collaboration, Technical Documentation, Problem Solving

WORK EXPERIENCE

DATA ENGINEER INTERN | flickDONE | LINK — **June 25- August 25**

- Optimized Python data pipelines processing 50,000+ daily records, improving machine learning feature reliability by 30%.
- Engineered SQL schemas and indexing strategies enabling scalable data pipelines and efficient retrieval for AI applications.
- Developed automated validation pipelines ensuring structured and unstructured dataset integrity for machine learning.
- Collaborated with engineering teams deploying scalable AI infrastructure pipelines supporting machine learning evaluation.
- Implemented Docker-based CI/CD pipelines with Git version control enabling reproducible ML deployments.

PROJECTS

Generative AI Product Recommender System | LINK — **June 25- August 2025**

- Developed Python recommendation engine using embeddings and machine learning models for personalization.
- Implemented similarity search using embeddings and vector retrieval techniques improving recommendation accuracy.
- Integrated Large Language Models with retrieval augmented generation pipelines enabling contextual recommendations.
- Deployed scalable backend APIs using Docker, Git workflows, and CI/CD improving reliability of production AI applications.

Rails TODO App | LINK — **September 24- November 2024**

- Developed full-stack task management platform using Ruby on Rails MVC architecture with responsive SCSS interface.
- Implemented CRUD workflows, authentication systems, and SQL queries supporting scalable backend logic.
- Optimized database performance through indexing strategies improving query efficiency and application responsiveness.
- Ensured maintainable architecture through modular code, Git version control, and structured Agile development workflows.

Morning Market Brief | LINK — **June 24- August 2024**

- Developed cloud-based analytics system generating automated financial insights using Python data processing pipelines.
- Built ETL workflows handling structured and unstructured datasets supporting downstream machine learning analytics tasks.
- Implemented REST API automation testing ensuring reliable data ingestion and scalable analytics infrastructure.
- Integrated Docker CI/CD pipelines enabling reproducible deployments and scalable backend services.

RESEARCH AND PUBLICATIONS

Multi-Stage Ensemble for Driver Fatigue Recognition | IEEE, 2025

- Developed PyTorch ensemble deep learning pipeline using audio embeddings for fatigue detection.
- Improved feature extraction pipelines optimizing preprocessing efficiency and reducing model inference time by 20%.
- Applied hyperparameter tuning, reproducible experiments, and evaluation workflows improving ML performance.

CERTIFICATES

Python, Data Science & ML (Cipherschools) | CERTIFICATE — **August 2024**

- Mastered Python programming fundamentals including object-oriented design and machine learning development practices.
- Applied machine learning algorithms, feature engineering, and model evaluation techniques on real-world datasets.

Generative AI with LLMs (Coursera) | CERTIFICATE — **March 2024**

- Learned transformer architectures, embeddings, and NLP fundamentals underlying Large Language Models.
- Built generative AI applications for summarization, semantic understanding, and intelligent workflow automation.